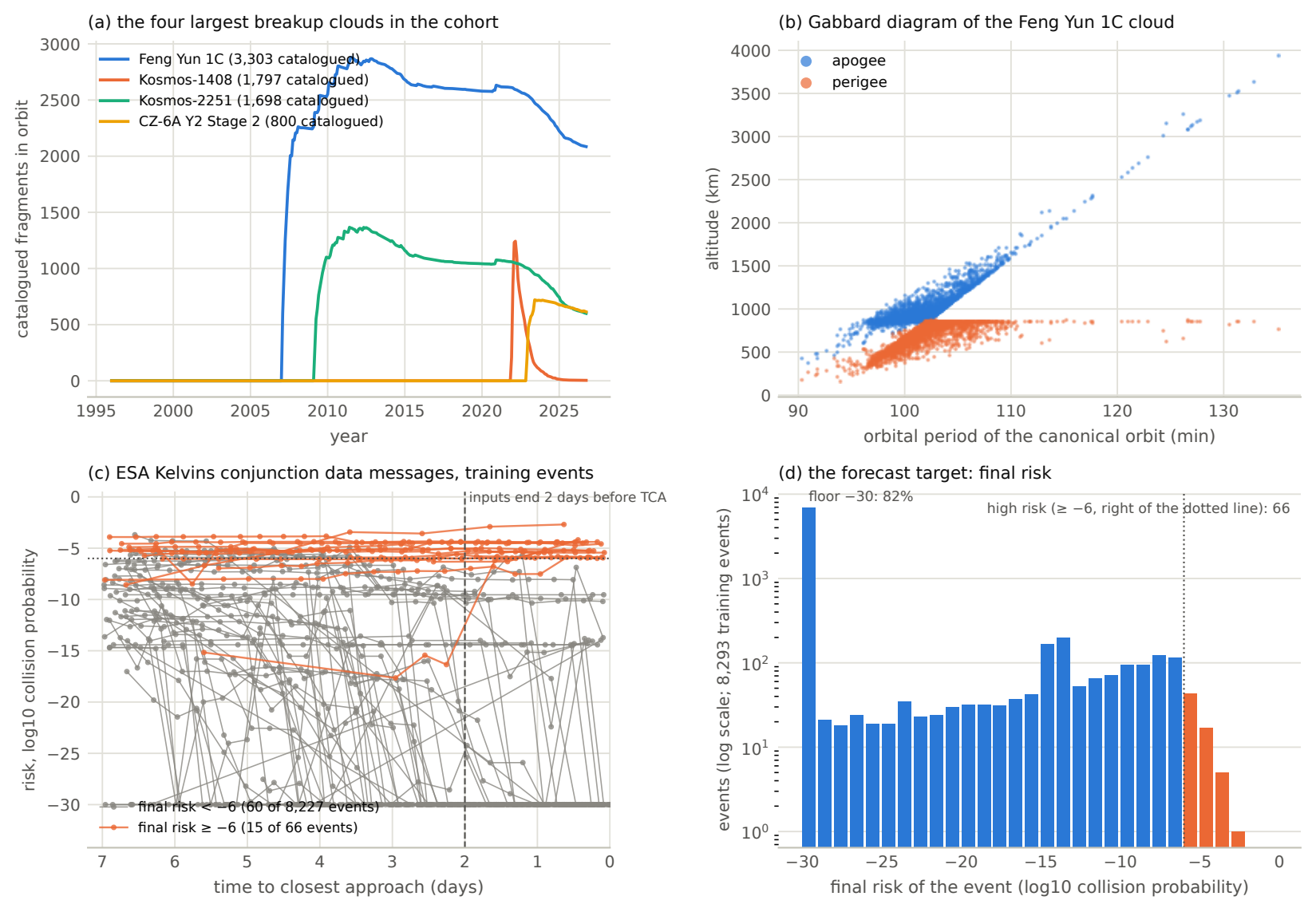

(a) the four largest breakup clouds in the cohort
Feng Yun 1C (3,303 catalogued)
Kosmos-1408 (1,797 catalogued)
Kosmos-2251 (1,698 catalogued)
CZ-6A Y2 Stage 2 (800 catalogued)
catalogued fragments in orbit
year
(b) Gabbard diagram of the Feng Yun 1C cloud
apogee
perigee
altitude (km)
orbital period of the canonical orbit (min)
(c) ESA Kelvins conjunction data messages, training events
inputs end 2 days before TCA
risk, log10 collision probability
final risk < −6 (60 of 8,227 events)
final risk ≥ −6 (15 of 66 events)
time to closest approach (days)
(d) the forecast target: final risk
floor −30: 82%
high risk (≥ −6, right of the dotted line): 66
events (log scale; 8,293 training events)
final risk of the event (log10 collision probability)